=== UNIT 1: WATER TECHNOLOGY, HARDNESS & SOFTENING PROCESSES ===

>> 1.1 Sources, Impurities and Hardness of Water

Water is the primary solvent, heat transfer fluid, and reaction medium in industrial biotechnology processes.

* Types of Hardness:

1. Temporary (Carbonate) Hardness: Caused by dissolved bicarbonates of calcium and magnesium [Ca(HCO3)2, Mg(HCO3)2]. Readily precipitated by boiling:
Ca(HCO3)2 --[Heat]--> CaCO3 (s) + H2O + CO2 (g).
Mg(HCO3)2 --[Heat]--> Mg(OH)2 (s) + 2 CO2 (g).
2. Permanent (Non-Carbonate) Hardness: Caused by dissolved chlorides, sulfates, and nitrates of calcium, magnesium, and heavy metals [CaCl2, MgCl2, CaSO4, MgSO4]. Cannot be removed by boiling; requires chemical precipitation or ion exchange.

* Units of Hardness: Standardized to equivalent parts of Calcium Carbonate (CaCO3, Mol Wt = 100, Eq Wt = 50):
- Parts per million (ppm): 1 ppm = 1 part of CaCO3 equivalent hardness per 10^6 parts of water.
- Milligrams per liter (mg/L): 1 mg/L = 1 mg of CaCO3 equivalent hardness per liter of water. (1 ppm = 1 mg/L).
- Degree Clark (deg Cl): 1 part CaCO3 per 70,000 parts water. (1 ppm = 0.07 deg Cl).
- Degree French (deg Fr): 1 part CaCO3 per 100,000 parts water. (1 ppm = 0.10 deg Fr).

* Equivalence Formula:
CaCO3 equivalent = [Mass of Hardness Substance (mg/L) * 50] / [Equivalent Weight of Substance].

>> 1.2 EDTA Complexometric Titration Method

* Principle: Disodium salt of ethylenediaminetetraacetic acid (Na2EDTA) forms stable, soluble 1:1 hexadentate chelate complexes with Ca2+ and Mg2+ ions at pH 10.0 (buffered with NH4OH + NH4Cl).
* Indicator: Eriochrome Black T (EBT). At pH 10, free EBT is blue. When added to hard water, it binds a small fraction of metal ions forming an unstable wine-red complex:
[M2+ + EBT] -> [M-EBT Complex] (Wine Red).
* Titration: As EDTA is added, it sequesters free Ca2+ and Mg2+ ions. At the equivalence point, EDTA displaces EBT from the wine-red complex:
[M-EBT Complex] (Wine Red) + EDTA -> [M-EDTA Chelate] (Colorless) + Free EBT (Blue).
Color change at endpoint: Sharp transition from WINE RED to SKY BLUE.

>> 1.3 Boiler Troubles and High-Pressure Mitigation

* Scales and Sludges: Scales are hard, adherent, crystalline crusts (CaSO4, CaCO3, CaSiO3) coating boiler tubes, causing localized overheating and tube blowout. Sludges are soft, slimy, non-adherent precipitates (MgCO3, MgCl2) settling in low-flow areas.
* Caustic Embrittlement: In high-pressure boilers, residual Na2CO3 hydrolyzes: Na2CO3 + H2O -> 2 NaOH + CO2. Concentrated NaOH penetrates hairline stress cracks in riveted boiler joints, dissolving protective Fe3O4: Fe + 2 NaOH -> Na2FeO2 + H2. Leaves steel brittle and prone to catastrophic failure. Prevented by adding sodium phosphate or tannin.
* Priming and Foaming: Priming is carryover of water droplets in steam; Foaming is persistent bubble formation on water surface caused by dissolved oils/organics. Prevented by anti-foaming agents (castor oil, polyamides).

>> 1.4 Modern Water Softening Technologies

* Zeolite / Permutit Process: Microporous hydrated sodium aluminosilicate (Na2O.Al2O3.xSiO2.yH2O, denoted Na2Ze). Hard water percolates through zeolite bed: Na2Ze + Ca(HCO3)2 -> CaZe + 2 NaHCO3; Na2Ze + MgSO4 -> MgZe + 2 Na2SO4. Exhausted bed regenerated with 10% brine: CaZe + 2 NaCl -> Na2Ze + CaCl2 (eluted to drain). Residual hardness: ~10-15 ppm. Cannot treat acidic or turbid water.
* Ion-Exchange (Demineralization) Process: Complete removal of all ions:
- Cation Exchanger (RH+): 2 RH + Ca2+ -> R2Ca + 2 H+. Regenerated with dilute HCl/H2SO4.
- Anion Exchanger (ROH-): 2 R'OH + SO4^2- -> R'2SO4 + 2 OH-. Regenerated with dilute NaOH.
- Effluent H+ and OH- neutralize: H+ + OH- -> H2O. Residual hardness: < 2 ppm (Deionized pure water).
* Reverse Osmosis (RO): Applying hydrostatic pressure exceeding the osmotic pressure (Delta P > Pi) across a semi-permeable thin-film composite membrane (polyamide/cellulose acetate) forces pure water against the concentration gradient, rejecting 99% of dissolved salts.

[SOLVED PROBLEM 1: Comprehensive Calculation of Water Hardness from Mineral Analysis]

Problem: A water sample from an industrial biotechnology supply contains the following dissolved mineral salts:
Ca(HCO3)2 = 16.2 mg/L, Mg(HCO3)2 = 14.6 mg/L, CaSO4 = 13.6 mg/L,
MgCl2 = 9.5 mg/L, MgSO4 = 12.0 mg/L, NaCl = 58.5 mg/L, SiO2 = 15.0 mg/L.
Calculate: (1) Temporary Hardness in ppm, (2) Permanent Hardness in ppm, (3) Total Hardness in ppm and French degrees.
Solution Step 1: Multiplication factor table (Equiv Wt of CaCO3 = 50):
- Ca(HCO3)2: Mol Wt = 162, Eq Wt = 81. Factor = 50 / 81 = 100 / 162.
CaCO3 equivalent = 16.2 * (50 / 81) = 10.0 mg/L (ppm).

- Mg(HCO3)2: Mol Wt = 146, Eq Wt = 73. Factor = 50 / 73 = 100 / 146.
CaCO3 equivalent = 14.6 * (50 / 73) = 10.0 mg/L (ppm).
- CaSO4: Mol Wt = 136, Eq Wt = 68. Factor = 50 / 68 = 100 / 136.
CaCO3 equivalent = 13.6 * (50 / 68) = 10.0 mg/L (ppm).
- MgCl2: Mol Wt = 95, Eq Wt = 47.5. Factor = 50 / 47.5 = 100 / 95.
CaCO3 equivalent = 9.5 * (50 / 47.5) = 10.0 mg/L (ppm).
- MgSO4: Mol Wt = 120, Eq Wt = 60. Factor = 50 / 60 = 100 / 120.
CaCO3 equivalent = 12.0 * (50 / 60) = 10.0 mg/L (ppm).
- Note: NaCl and SiO2 do not cause hardness (non-hardness impurities).
Solution Step 2: Temporary Hardness (Sum of Bicarbonates):
Temporary Hardness = Equivalent [Ca(HCO3)2 + Mg(HCO3)2] = 10.0 + 10.0 = 20.0 ppm.
Solution Step 3: Permanent Hardness (Sum of Sulfates and Chlorides):
Permanent Hardness = Equivalent [CaSO4 + MgCl2 + MgSO4] = 10.0 + 10.0 + 10.0 = 30.0 ppm.
Solution Step 4: Total Hardness:
Total Hardness = Temporary + Permanent = 20.0 + 30.0 = 50.0 ppm (mg/L).
Solution Step 5: Convert to Degree French (1 ppm = 0.10 deg Fr):
Total Hardness in deg Fr = 50.0 * 0.10 = 5.0 deg French.
Conclusion: Water sample has 20 ppm temporary, 30 ppm permanent, and 50 ppm total hardness.

[SOLVED PROBLEM 2: EDTA Titration Quantitative Analysis]

Problem: 50 mL of standard hard water containing 1.0 mg of pure CaCO3 per mL required 20.0 mL of EDTA solution for titration. 100 mL of an unknown hard water sample required 30.0 mL of the same EDTA solution. 100 mL of the same water sample after boiling and filtering required 10.0 mL of EDTA solution. Calculate: (1) Total hardness, (2) Permanent hardness, (3) Temporary hardness in ppm.
Solution Step 1: Standardize EDTA Solution:
50 mL of standard water = 50 * 1.0 mg = 50.0 mg CaCO3 equivalent.
20.0 mL EDTA = 50.0 mg CaCO3 => 1.0 mL EDTA = 50.0 / 20.0 = 2.50 mg CaCO3 equivalent.
Solution Step 2: Total Hardness of Unknown Sample:
100 mL water required 30.0 mL EDTA = 30.0 * 2.50 = 75.0 mg CaCO3 equivalent.
In 1,000 mL (1 Liter) of water = 75.0 * (1000 / 100) = 750.0 mg/L.
Total Hardness = 750.0 ppm.
Solution Step 3: Permanent Hardness (Boiled Water):
100 mL boiled water required 10.0 mL EDTA = 10.0 * 2.50 = 25.0 mg CaCO3 equivalent.
In 1,000 mL of water = 25.0 * (1000 / 100) = 250.0 mg/L.
Permanent Hardness = 250.0 ppm.
Solution Step 4: Temporary Hardness:
Temporary Hardness = Total Hardness - Permanent Hardness = 750.0 - 250.0 = 500.0 ppm.
Final Results: Total = 750 ppm | Permanent = 250 ppm | Temporary = 500 ppm.

[SOLVED PROBLEM 3: Zeolite Softener Capacity and Salt Regeneration Calculation]

Problem: A zeolite water softener bed completely softened 40,000 liters of hard water containing 350 ppm of total hardness before exhaustion. Calculate the mass of sodium chloride (NaCl) in kilograms required to regenerate the exhausted zeolite bed using 10% brine solution.

Solution Step 1: Total CaCO3 Equivalent Hardness Removed:

Total water volume = 40,000 Liters.

Hardness = 350 mg/L = 350 g / 1000 L.

Total CaCO3 hardness removed = 40,000 L * 350 mg/L = 14,000,000 mg = 14.0 kg CaCO3 equivalent.

Solution Step 2: Stoichiometric NaCl Regeneration Relationship:

Regeneration reaction: CaZe + 2 NaCl -> Na2Ze + CaCl2.

100 g CaCO3 equivalent requires 2 * 58.5 g = 117.0 g NaCl.

Solution Step 3: Calculate Required Mass of NaCl:

Mass of pure NaCl = [Total CaCO3 equivalent (kg) * 117.0] / 100.0

Mass of NaCl = [14.0 kg * 117.0] / 100.0 = 16.38 kg of pure NaCl.

If commercial salt is 95% pure:

Commercial Salt Required = 16.38 / 0.95 = 17.24 kg.

[SOLVED PROBLEM 4: Reverse Osmosis Operating Pressure Calculation]

Problem: Calculate the minimum theoretical hydrostatic pressure (P_min) required to desalinate brackish water containing 5,850 mg/L of dissolved NaCl at 25 deg C (298 K) via Reverse Osmosis. (Van't Hoff factor i = 2 for NaCl; gas constant R = 0.0821 L*atm/mol*K).

Solution Step 1: Compute Molar Concentration of NaCl:

Molar mass of NaCl = 58.44 g/mol.

Concentration C = 5.850 g/L / 58.44 g/mol = 0.1001 mol/L.

Solution Step 2: Calculate Osmotic Pressure Pi via Van't Hoff Equation:

Pi = i * C * R * T

Pi = 2 * (0.1001 mol/L) * (0.0821 L*atm/mol*K) * (298 K)

Pi = 2 * 0.1001 * 24.4658 = 4.898 atm.

Solution Step 3: Convert to SI Units (1 atm = 101,325 Pa = 1.01325 bar):

Pi = 4.898 * 1.01325 bar = 4.96 bar ~ 5.0 bar.

Conclusion: The applied hydrostatic pump pressure must strictly exceed 4.96 bar (typically operated at 15-20 bar in practice to maintain adequate permeate flux).

>> 1.5 University Review & Examination Questions

* Q1: Explain the chemical principles of the EDTA titration method for determining water hardness, giving structures and indicator color change mechanisms.

Answer: EDTA (ethylenediaminetetraacetic acid) is a hexadentate chelating ligand possessing two amino nitrogens and four carboxylate oxygens that wrap around metal cations in a stable octahedral cage. The titration is conducted at pH 10.0 using NH4Cl-NH4OH buffer to ensure complete deprotonation of EDTA to EDTA^4- while preventing precipitation of Mg(OH)2. Eriochrome Black T (EBT) indicator is added, binding Ca2+/Mg2+ to form an unstable wine-red complex: M2+ + EBT -> [M-EBT] (wine-red). During titration, free EDTA preferentially chelates free Ca2+ and Mg2+ to form colorless, highly stable [M-EDTA]^2- complexes (formation constant log K_f ~ 10.7 for Ca). At the equivalence point, EDTA displaces EBT from the metal-indicator complex. Free, uncomplexed EBT is liberated into the alkaline buffer, producing a sharp, irreversible transition from WINE RED to SKY BLUE.

* Q2: Compare Zeolite process with Ion-Exchange Demineralization process across chemistry, residual hardness, and operational constraints.

Answer: (1) Chemistry: Zeolite uses inorganic hydrated sodium aluminosilicates (Na2Ze) that exchange only cations (Ca2+, Mg2+) for Na+; anions remain in water. Ion-Exchange uses separate synthetic organic polystyrene resins: cation resin (R-SO3H) exchanges all cations for H+, and anion resin (R-NMe3OH) exchanges all anions for OH-. (2) Residual Hardness: Zeolite leaves 10-15 ppm residual hardness; Ion-Exchange produces completely demineralized water with < 2 ppm hardness (electrical conductivity < 0.1 microS/cm). (3) Treated Water Composition: Zeolite water contains high NaHCO3, causing caustic embrittlement in high-pressure boilers. Ion-Exchange effluent contains zero dissolved solids. (4) Constraints: Zeolite cannot treat acidic or turbid water (acids destroy the aluminosilicate cage); Ion-Exchange resins handle acidic feed streams safely.

>> 1.6 Key Takeaways and Revision Points

* Hardness is standardized to CaCO3 equivalents: Equiv = [Mass * 50] / Eq Wt.
* EDTA titration at pH 10 uses EBT indicator, transitioning from wine-red to sky blue.
* Boiler troubles include scale/sludge, caustic embrittlement (NaOH attack), priming, and foaming.
* Ion-exchange demineralization removes all cations and anions, yielding ultra-pure water (<2 ppm).